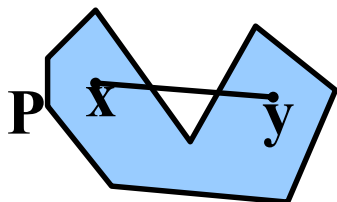


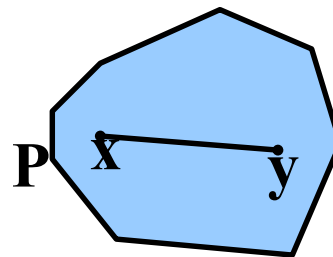
Convex Hull

Convex vs. Concave

- A polygon P is **convex** if for every pair of points x and y in P , the line xy is also in P ; otherwise, it is called **concave**.



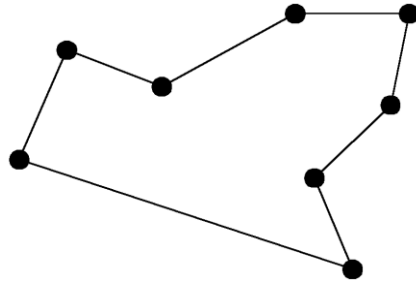
concave



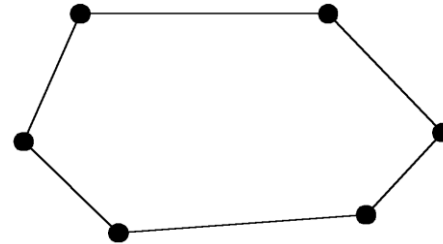
convex

The convex hull problem

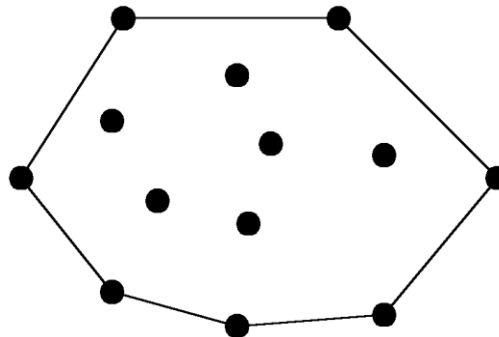
concave polygon:



convex polygon:



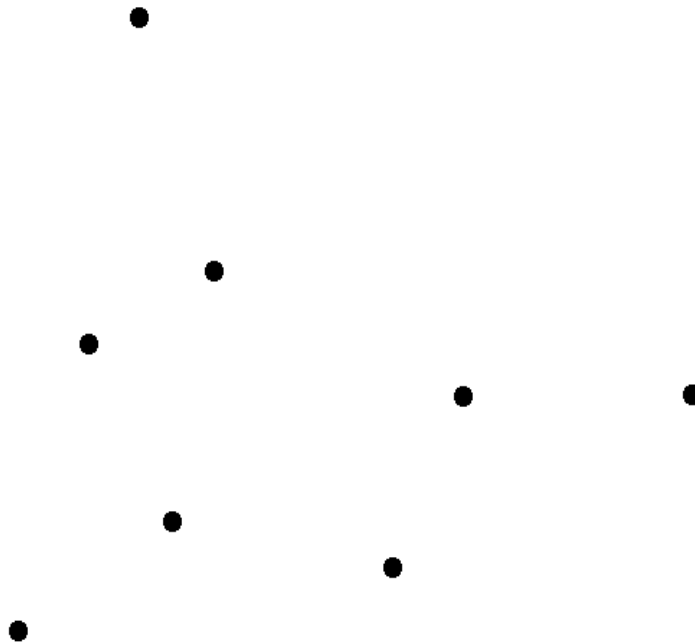
- The convex hull of a set of planar points is the smallest convex polygon containing all of the points.



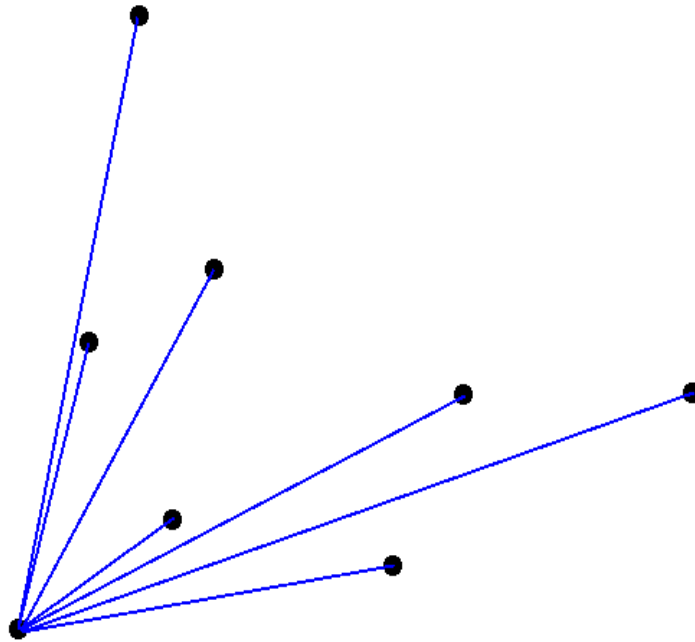
Graham's Scan

- Start at point guaranteed to be on the hull. (the point with the minimum y value)
- **Sort** remaining points by **polar angles** of vertices relative to the first point.
- Go through sorted points, keeping vertices of points that have **left turns** and dropping points that have **right turns**.

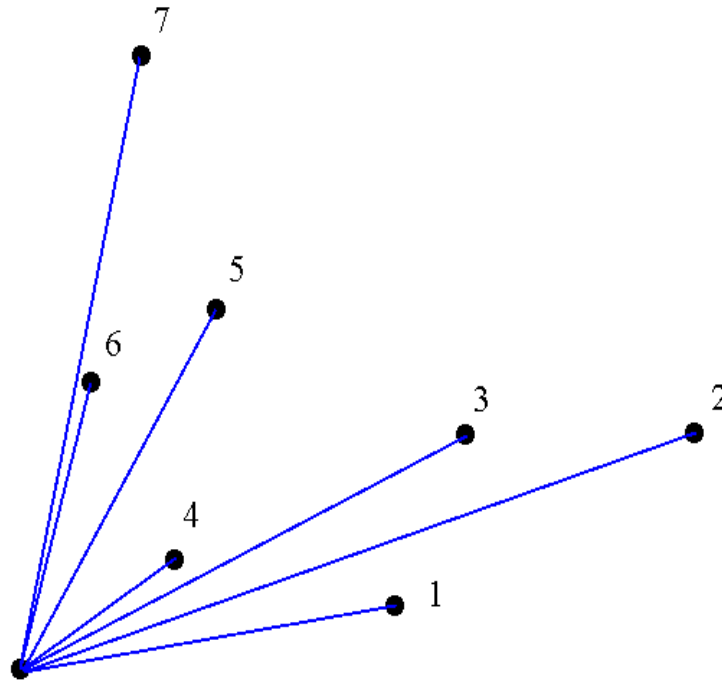
Graham's Scan



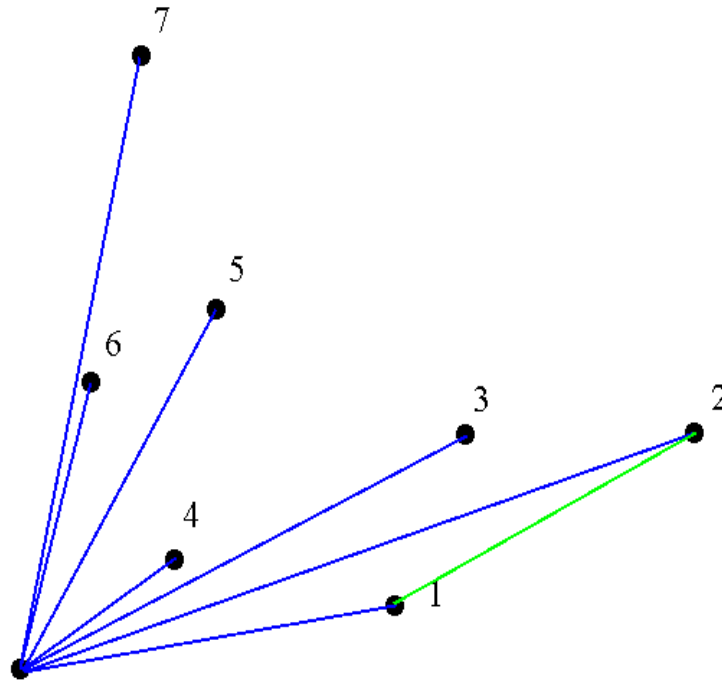
Graham's Scan



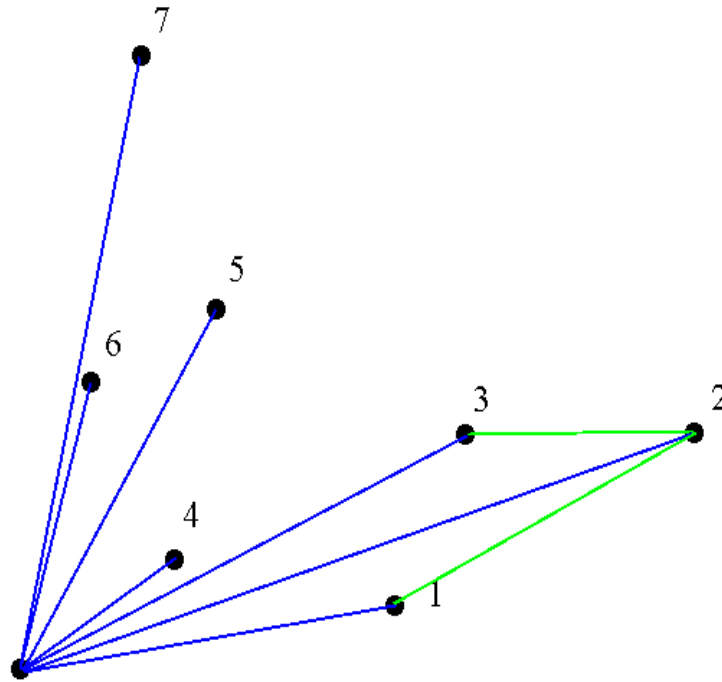
Graham's Scan



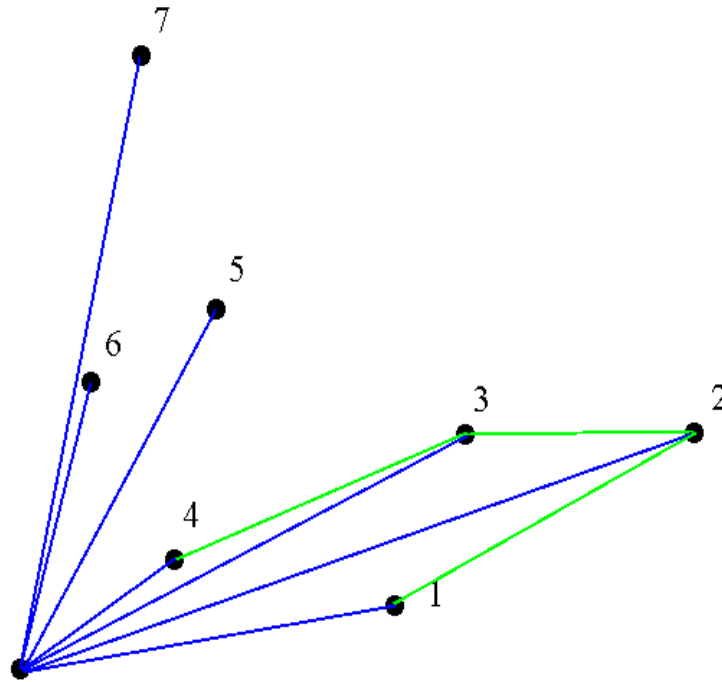
Graham's Scan



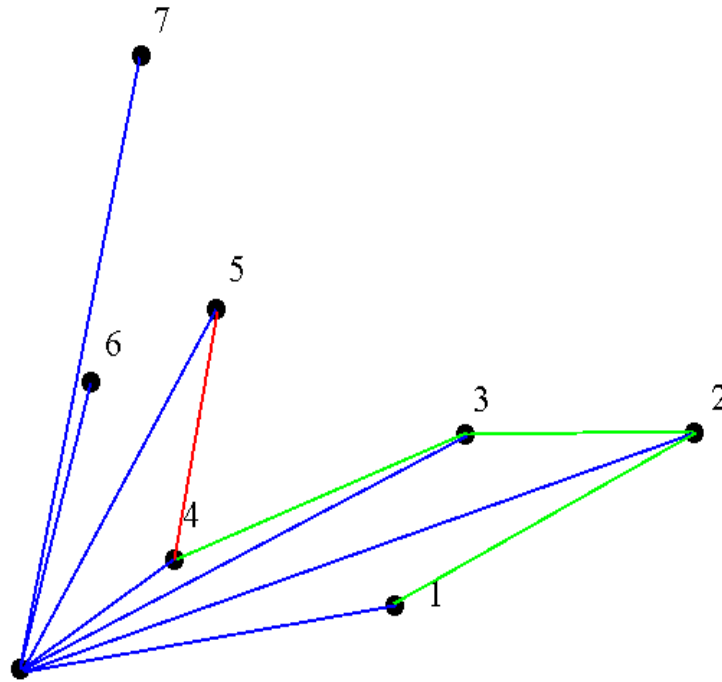
Graham's Scan



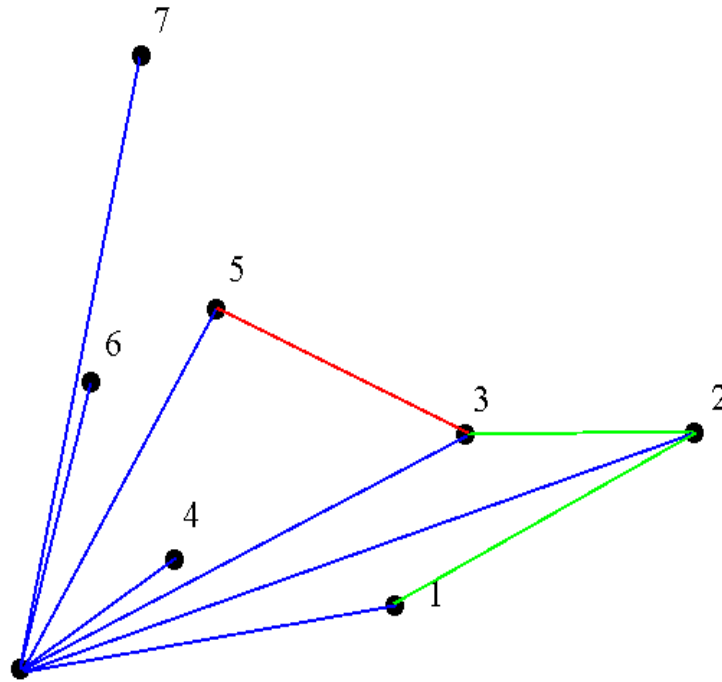
Graham's Scan



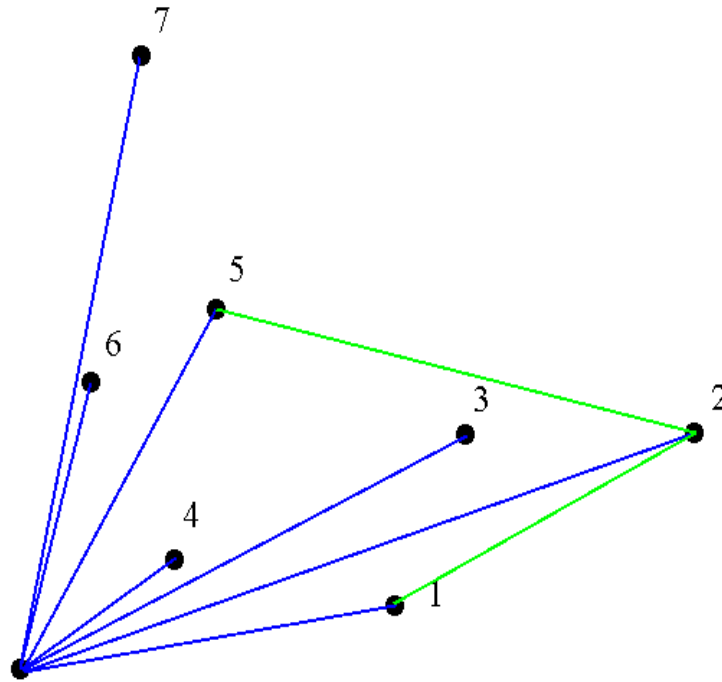
Graham's Scan



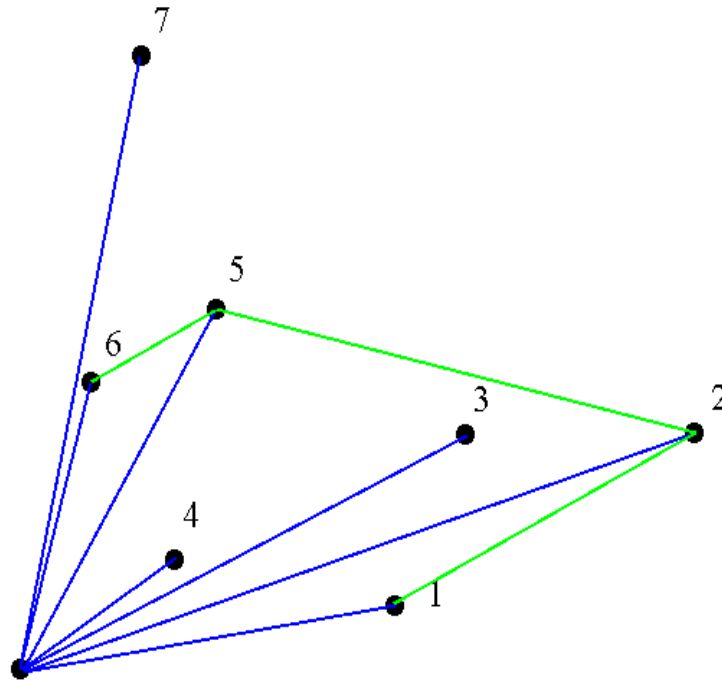
Graham's Scan



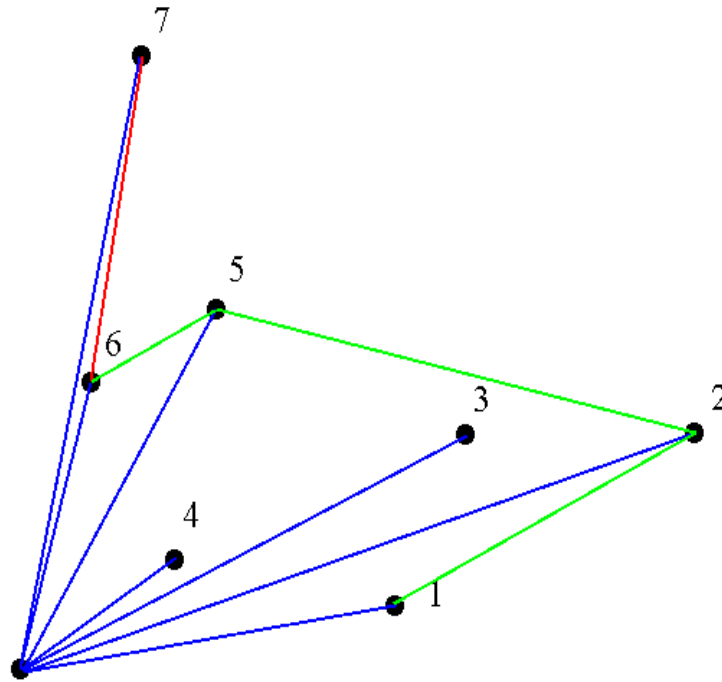
Graham's Scan



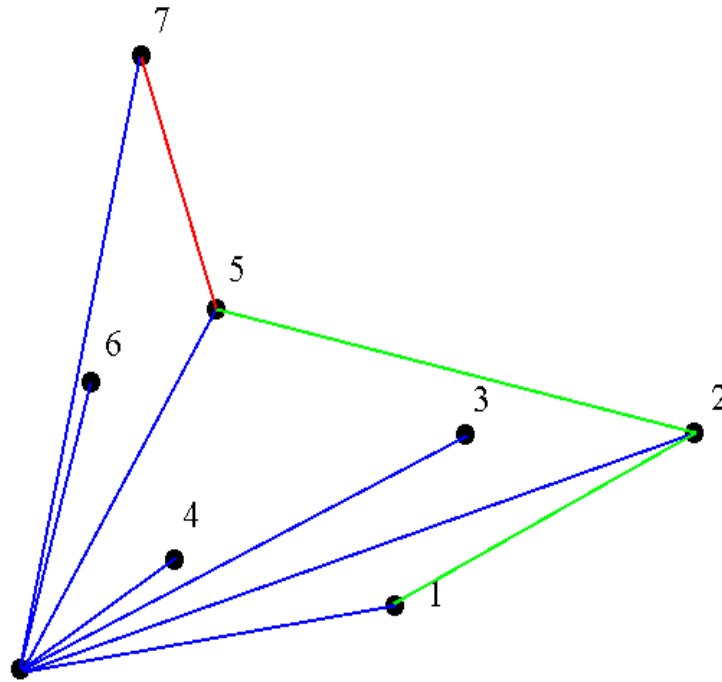
Graham's Scan



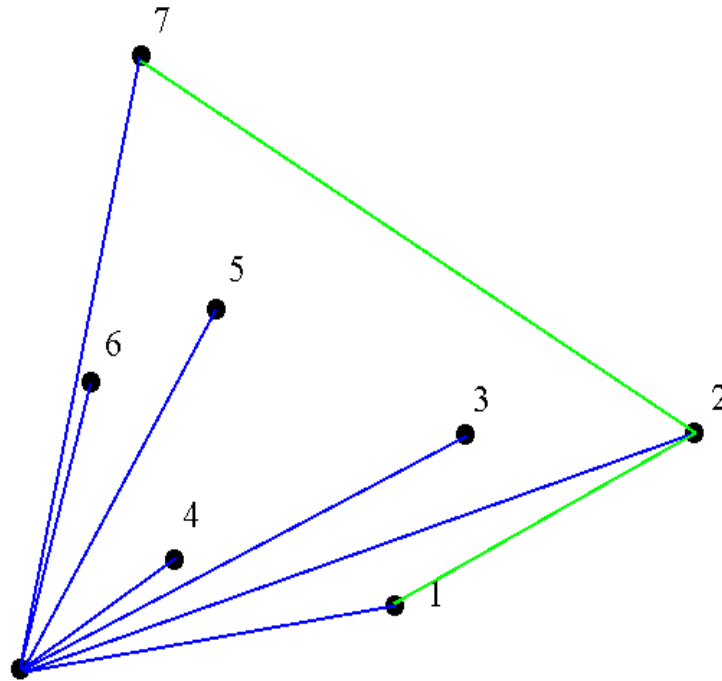
Graham's Scan



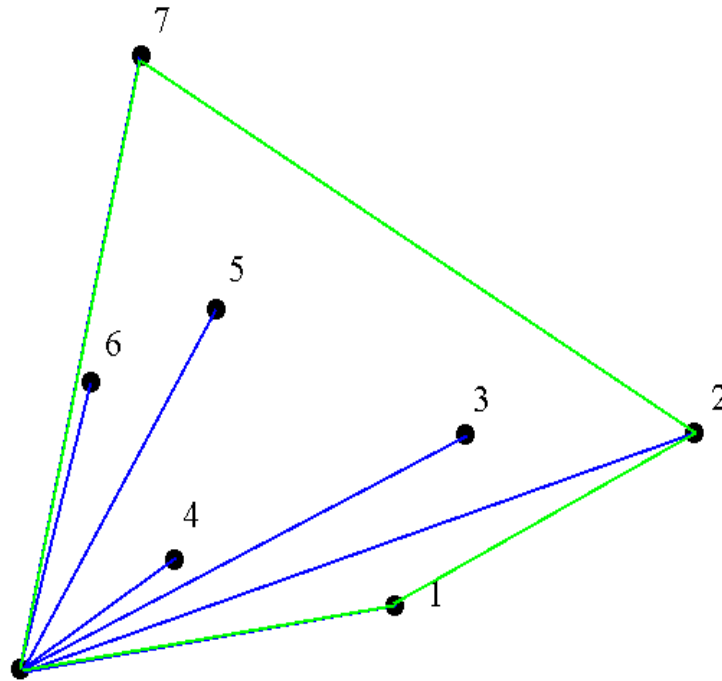
Graham's Scan



Graham's Scan



Graham's Scan



Graham's Runtime

- Graham's scan is $O(n \log n)$ due to initial sort of angles.

$O(n) + O(n \log n) + O(n)$
pick+sort+line draw

EXam ---> need to step to step showing

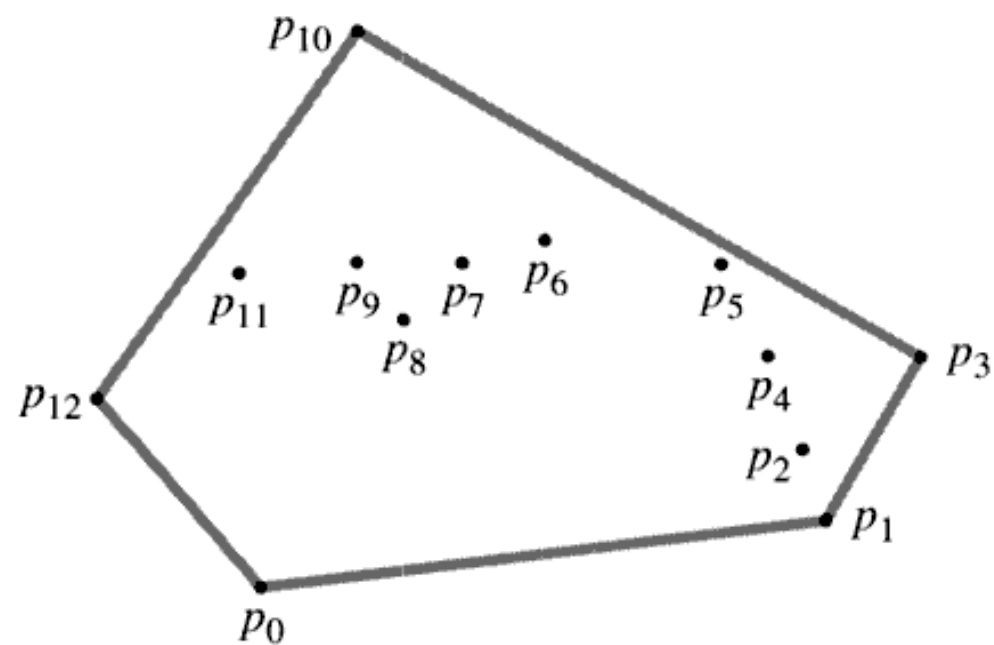
Detailed algorithm

GRAHAM-SCAN(Q)

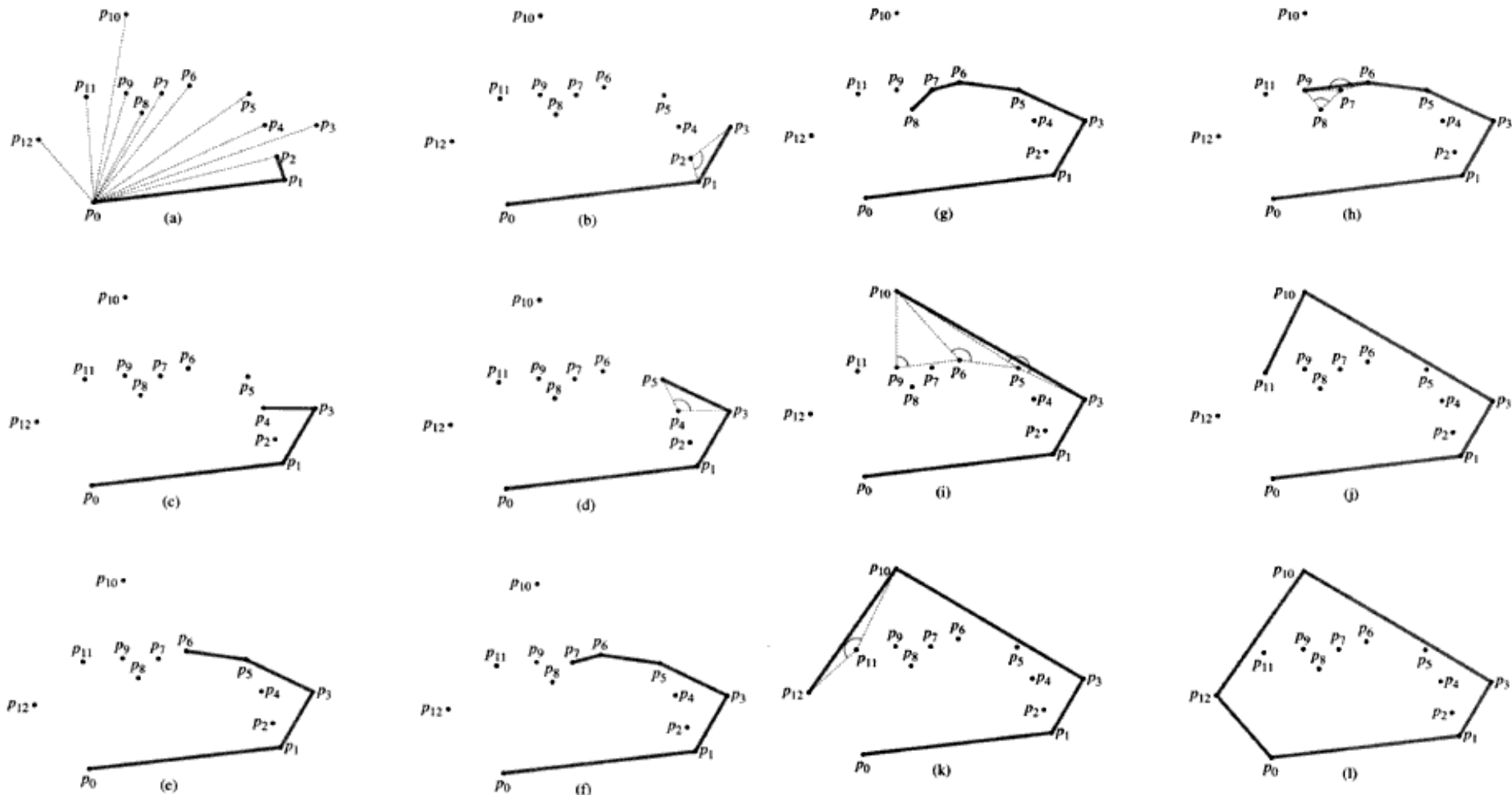
- ```

1 let p_0 be the point in Q with the minimum y -coordinate,
 or the leftmost such point in case of a tie
2 let $\langle p_1, p_2, \dots, p_m \rangle$ be the remaining points in Q ,
 sorted by polar angle in counterclockwise order around p_0
 (if more than one point has the same angle, remove all but
 the one that is farthest from p_0)
3 PUSH(p_0, S)
4 PUSH(p_1, S)
5 PUSH(p_2, S)
6 for $i \leftarrow 3$ to m
7 do while the angle formed by points NEXT-TO-TOP(S), TOP(S),
 and p_i makes a nonleft turn
8 do POP(S)
9 PUSH(p_i, S)
10 return S

```



**Figure 33.6** A set of points  $Q = \{p_0, p_1, \dots, p_{12}\}$  with its convex hull  $\text{CH}(Q)$  in gray.



**Figure 33.7** The execution of GRAHAM-SCAN on the set  $Q$  of Figure 33.6. The current convex hull contained in stack  $S$  is shown in gray at each step. (a) The sequence  $\langle p_1, p_2, \dots, p_{12} \rangle$  of points numbered in order of increasing polar angle relative to  $p_0$ , and the initial stack  $S$  containing  $p_0, p_1$ , and  $p_2$ . (b)–(k) Stack  $S$  after each iteration of the **for** loop of lines 6–9. Dashed lines show nonleft turns, which cause points to be popped from the stack. In part (h), for example, the right turn at angle  $\angle p_7 p_8 p_9$  causes  $p_8$  to be popped, and then the right turn at angle  $\angle p_6 p_7 p_9$  causes  $p_7$  to be popped. (l) The convex hull returned by the procedure, which matches that of Figure 33.6.

# Convex Hull by Divide-and-Conquer

---

- First, sort all points by their x coordinate.
  - (  $O(n \log n)$  time)
- Then divide and conquer:
  - Find the convex hull of the left half of points.
  - Find the convex hull of the right half of points.
  - Merge the two hulls into one.

# Convex Hull Pseudocode

---

```
//input: the number of points n, and
//an array of points S, sorted by x coord.
//output: the convex hull of the points in S.

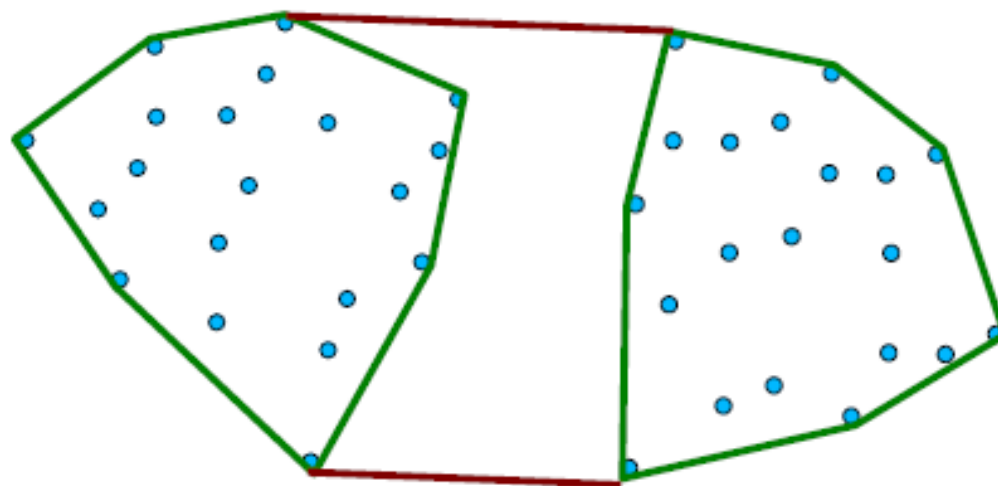
point[] findHullDC(int n, point S[]) {
 if (n > 5) {
 int h = floor(n/2);
 m = n-h;
 point LH[], RH[]; //left and right hulls
 LH = findHullDC(h, S[1..h]);
 RH = findHullDC(m, S[h+1..n]);
 return mergeHulls(LH.size(), RH.size(),
 LH, RH);
 } else {
 return Hull of S by exhaustive search;
 }
}
```



# Merging Hulls

---

- Big picture:
  - first find the lines that are upper tangent, and lower tangent to the two hulls (the two red lines)

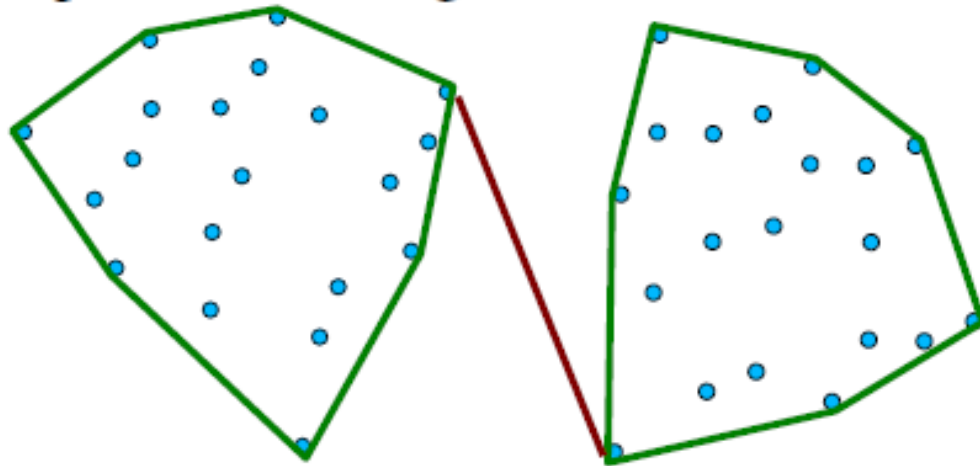


- Then remove the points that are cut off.

# Finding Tangent Lines

---

- Start with the rightmost point of the left hull, and the leftmost point of the right hull:

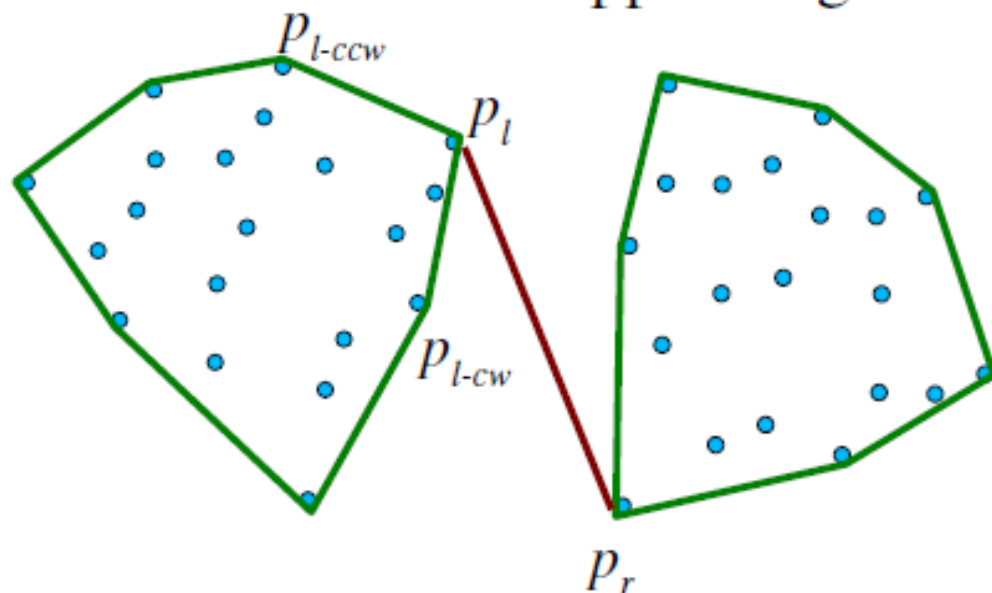


- While the line is not upper tangent to both left and right:
  - While the line is not upper tangent to the left, move to the next point (counter-clockwise).
  - While the line is not upper tangent to the right, move to the next point (clockwise).

# Checking Tangentness

---

- How can we tell if a line is upper tangent to the left hull?



- The pair of line segments  $\overline{p_r p_l}$  , and  $\overline{p_l p_{l-ccw}}$  should make a CCW turn at  $p_l$
- The same goes for  $\overline{p_r p_l}$  and  $\overline{p_l p_{l-cw}}$  .

# Finding the lower tangent in $O(n)$ time

a = rightmost point of A

b = leftmost point of B

while T=ab not lower tangent to both  
convex hulls of A and B do{

while T not lower tangent to  
convex hull of A do{

a=a-1

}

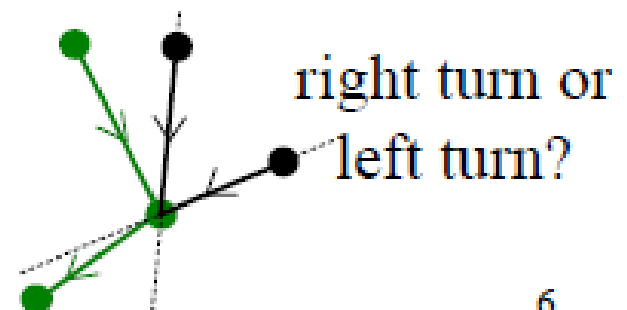
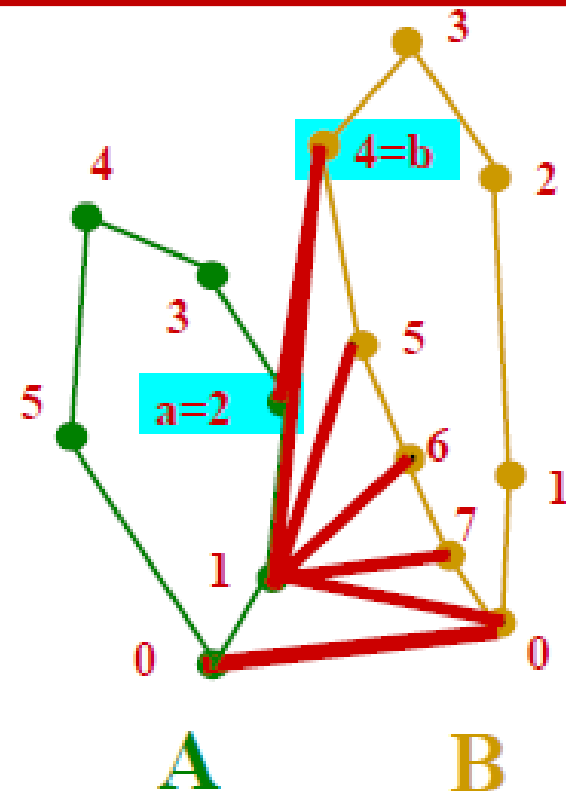
while T not lower tangent to  
convex hull of B do{

b=b+1

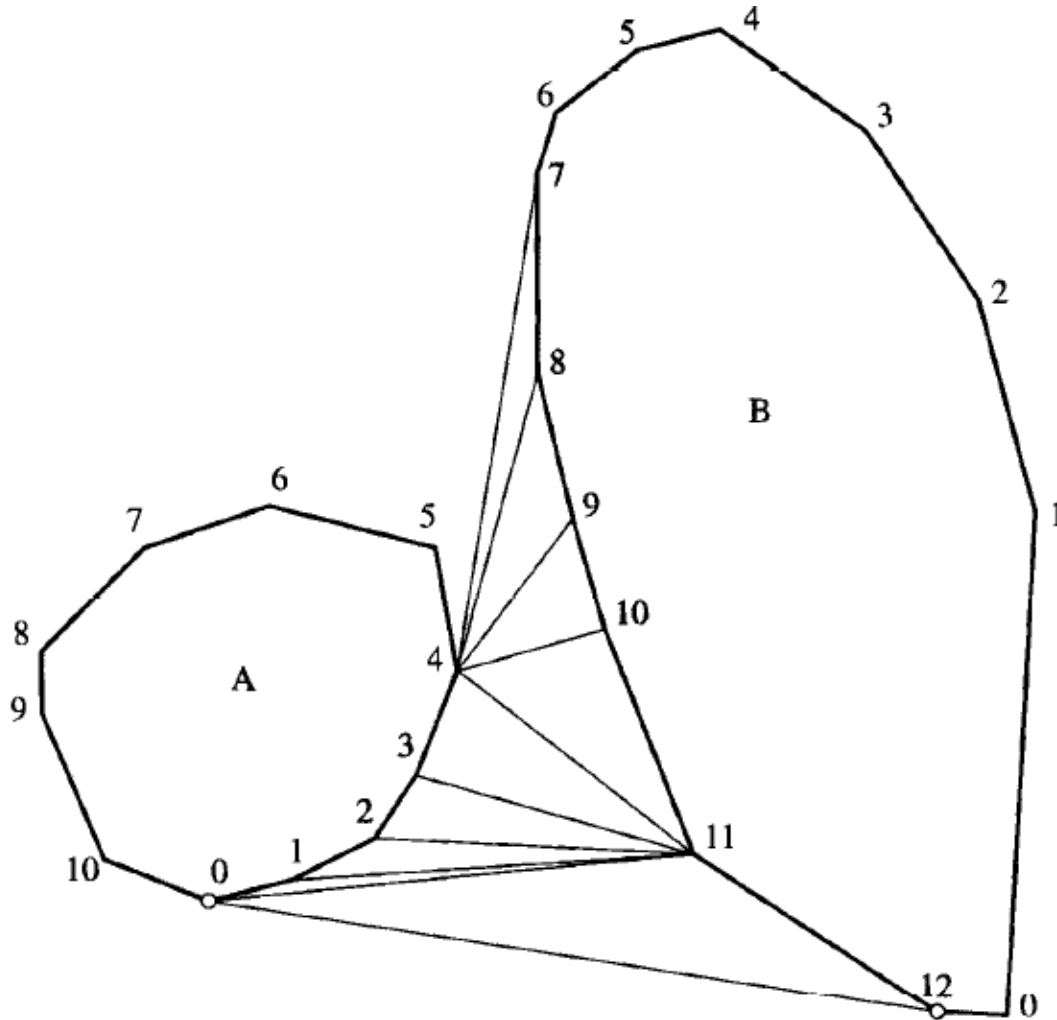
}

}

can be checked  
in constant time



# Lower Tangent Example



- Initially,  $T=(4, 7)$  is only a lower tangent for A. The A loop does not execute, but the B loop increments  $b$  to 11.
- But now  $T=(4, 11)$  is no longer a lower tangent for A, so the A loop decrements  $a$  to 0.
- $T=(0, 11)$  is not a lower tangent for B, so  $b$  is incremented to 12.
- $T=(0, 12)$  is a lower tangent for both A and B, and  $T$  is returned.

# Convex Hull: Runtime

|                                                                  |                         |
|------------------------------------------------------------------|-------------------------|
| • Preprocessing: sort the points by x-coordinate                 | $O(n \log n)$ just once |
| • Divide the set of points into two sets <b>A</b> and <b>B</b> : | $O(1)$                  |
| • <b>A</b> contains the left $\lfloor n/2 \rfloor$ points,       |                         |
| • <b>B</b> contains the right $\lceil n/2 \rceil$ points         |                         |
| • Recursively compute the convex hull of <b>A</b>                | $T(n/2)$                |
| • Recursively compute the convex hull of <b>B</b>                | $T(n/2)$                |
| • Merge the two convex hulls                                     | $O(n)$                  |

$$T(n) = 2 T(n/2) + cn$$

$$T(n) = O(n \log n)$$

---

Q&A